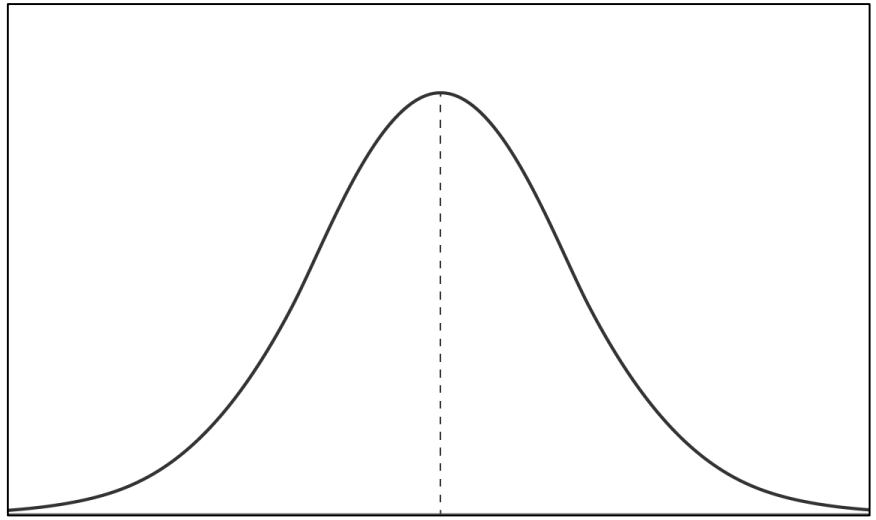
Probability Distribution (3)

Shuvongkar Sarkar

Normal Distribution

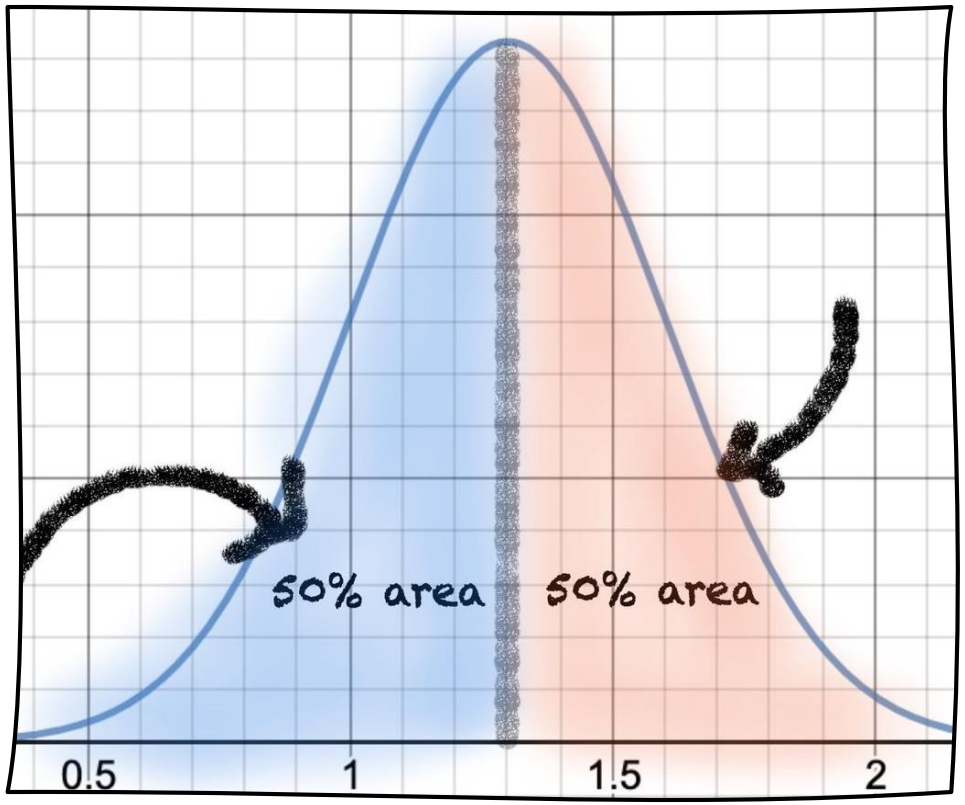
- Normal distribution is a probability distribution that is symmetric about the mean

- Also known as the Gaussian distribution



Properties Normal Distribution

1. Mean is the middle value and divides the area in half
2. Mean = Median = Mode
3. Symmetric and Mesokurtic
4. Bell-shaped curve



Normal Distribution

Let, X be a continuous random variable

Then the normal distribution function can be written as

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad ; -\infty < x < \infty$$

$$X \sim N(\mu, \sigma^2)$$

Normal Distribution

- Mean of normal distribution, $E(X) = \mu$
- Variance of normal distribution, $V(X) = \sigma^2$

Standard Normal Distribution

Let, $Z = \frac{X-\mu}{\sigma}$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \; ; -\infty < z < \infty$$

Mean, $E(Z) = E\left(\frac{X-\mu}{\sigma}\right) = \frac{E(X)-\mu}{\sigma} = \frac{\mu-\mu}{\sigma} = 0$

Variance, $V(Z) = V\left(\frac{X-\mu}{\sigma}\right) = \frac{V(X)}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1$

$Z \sim N(0,1)$

Standard Normal Distribution

- $X = 3, 11, 11, 11, 15, 15$

  a. Determine the standardize variable Z corresponding to X $Z = \frac{X - \mu}{\sigma} = \frac{X - 11}{4}$

  b. Find the $Z - score$ of an observed value of $X = 11$ $Z = \frac{X - \mu}{\sigma} = \frac{11 - 11}{4} = 0$

  c. Compute all possible $Z - scores$

  d. Find mean and standard deviation of variable Z.

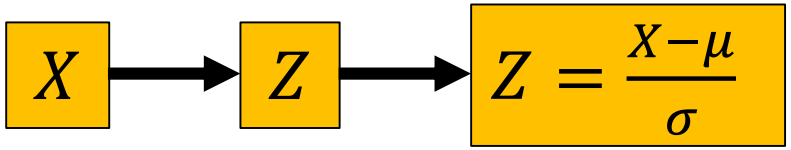
Probability Calculation

- There are two methods to compute the probabilities associated with a normal variable:

  a. Using Table (Z table)

  b. Using Calculator

Steps



- To find probabilities for a normal random variable X, we can transform the probability statement about X in terms of probability statement for Z

- Then calculate the probability using the standard normal distribution table/Z-table or Calculator

Example

- The number of viewers of UCL per week has a mean of 60 million with a standard deviation of 7 million. Assume that, the number of viewers of that show follows a normal distribution.

What is the probability that, next week's show will-

a. Have between 61 and 64 million viewers?
b. Have at least 53 million viewers?
c. Exceed 50 million viewers?

Example

Transform the probability statement in terms of Z

$$Z = \left(\frac{X-\mu}{\sigma}\right)$$

Let, X = Number of viewers of the show per week (in million)

Here, $X \sim N(\mu, \sigma^2)$; i.e., $X \sim N(60, 49)$

a. Have between 61 and 64 million viewers?

$P(61 \leq X \leq 64)$

$$P\left(\frac{61-\mu}{\sigma} \leq \frac{X-\mu}{\sigma}\right)$$

$P(a \leq X \leq b)$

$P(b) - P(a)$

$P(a < X < b)$

$$P\left(\frac{61-\mu}{\sigma} \leq Z \leq \frac{64-\mu}{\sigma}\right)$$

$$P\left(\frac{61-60}{7} \leq Z \leq \frac{64-60}{7}\right)$$

$P(0.14 \leq Z$

$P(0.57)$
$- P(0.14)$

???−???

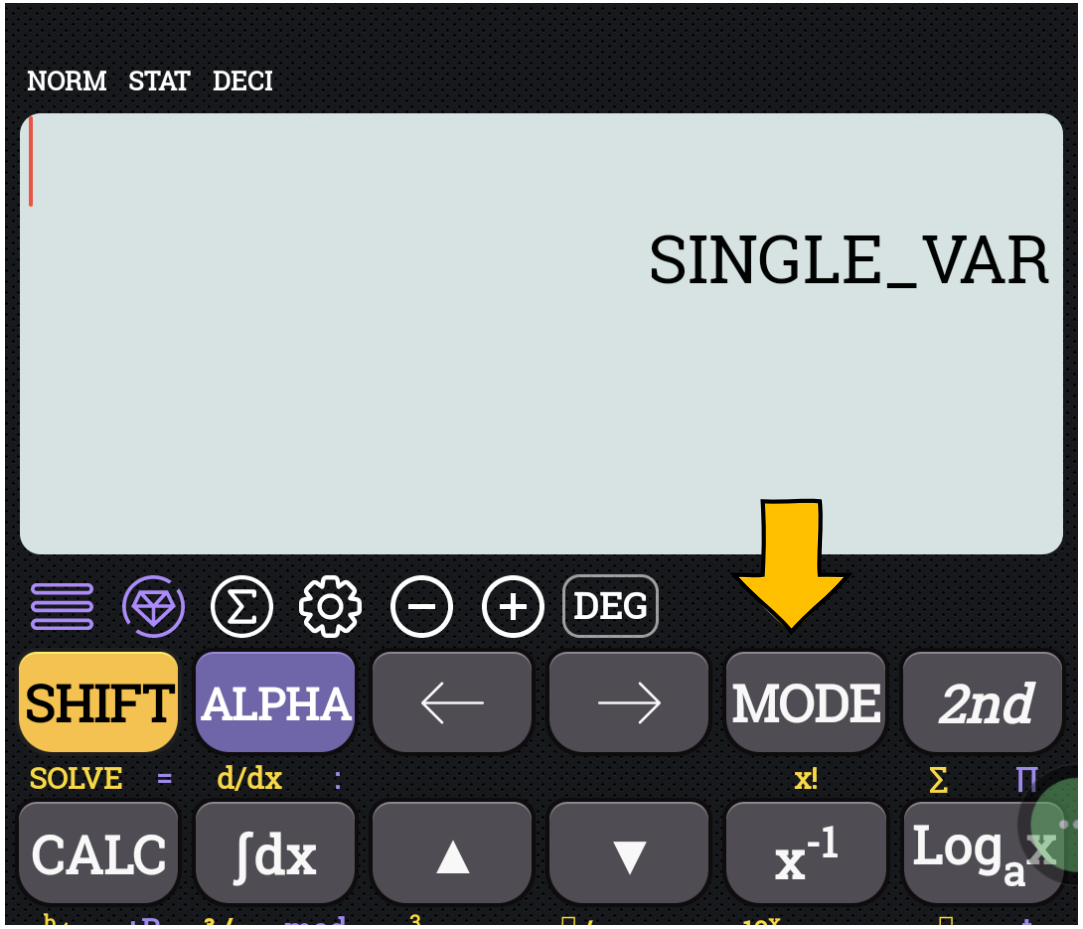
Z table

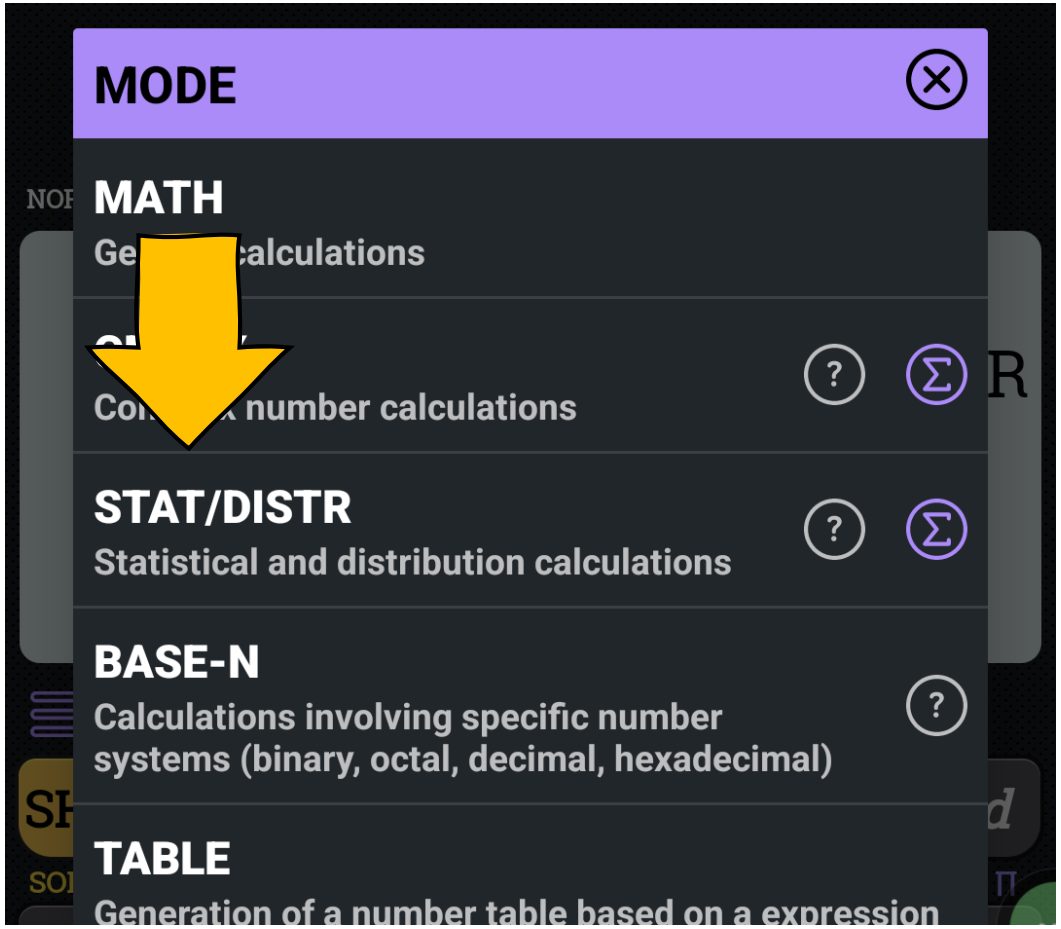
TABLE A

Standard normal probabilities (continued)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319

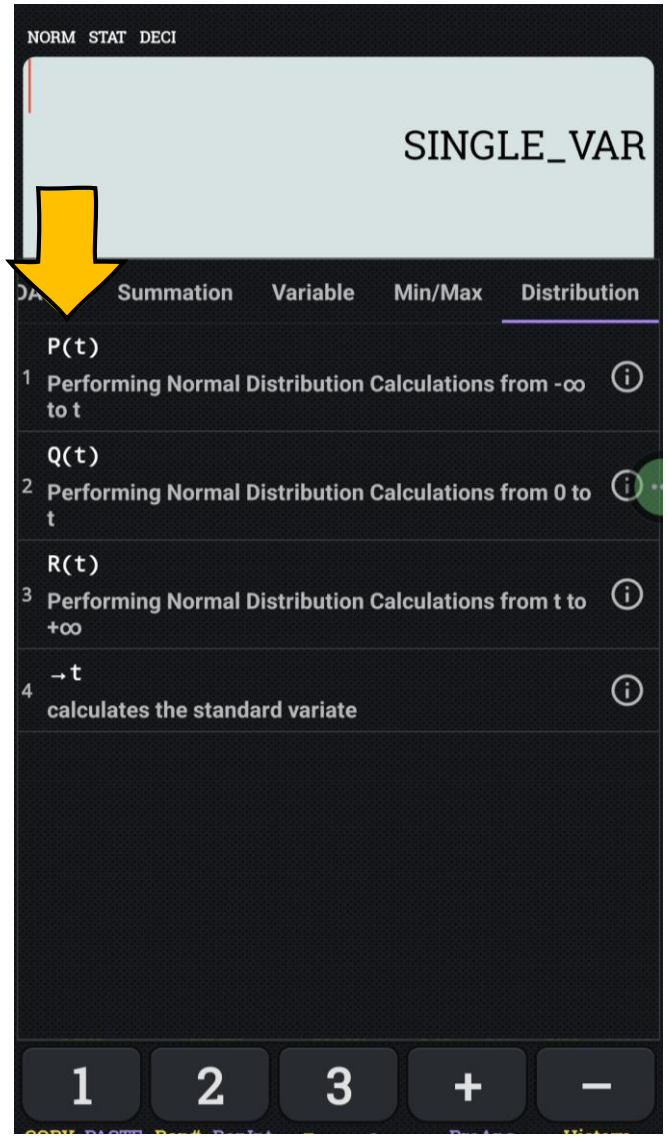
Calculator

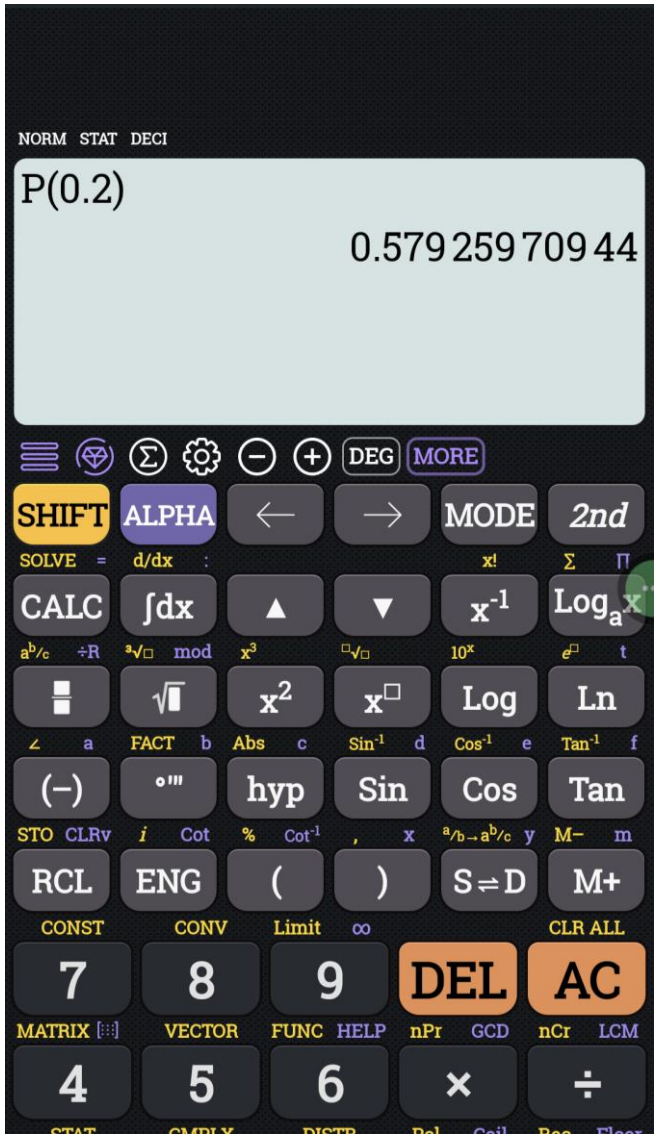

NORM STAT DECI
SINGLE_VAR
DEG
SHIFT
ALPHA
MODE
2nd
SOLVE = d/dx :
x! Σ Π
CALC
∫dx
x-1
Loga x


MODE
MATH
Ge calculations
number calculations
STAT/DISTR
Statistical and distribution calculations
BASE-N
Calculations involving specific number systems (binary, octal, decimal, hexadecimal)
TABLE
Generation of a number table based on a expression

Calculator


NORM STAT DECI
SINGLE_VAR
DEG
SHIFT
ALPHA
MODE
2nd
CALC
∫dx
Log
Ln
hyp
Sin
Cos
Tan
RCL
ENG
S⇌D
M+
CONST
CONV
Limit
CLR ALL
8
9
DEL
AC
VECTOR
FUNC
HELP
nPr
GCD
nCr
LCM
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STAT
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DISTR
Pol
Ceil
Rec
Floor
1
2
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COPY
PASTE
Ran#
RanInt
PreAns
History
0
.
Exp
Ans
=


NORM STAT DECI
SINGLE_VAR
Summation
Variable
Min/Max
Distribution
P(t)
1 Performing Normal Distribution Calculations from -∞ to t
Q(t)
2 Performing Normal Distribution Calculations from 0 to t
R(t)
3 Performing Normal Distribution Calculations from t to +∞
→t
4 calculates the standard variate
1
2
3
+
−


NORM STAT DECI
P(0.2)
0.579 259 709 44
DEG
MORE
SHIFT
ALPHA
MODE
2nd
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∫dx
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Video tutorial

CASIO 991 ES

CASIO 991 EX

Normal Distribution

Transform the probability statement in terms of Z

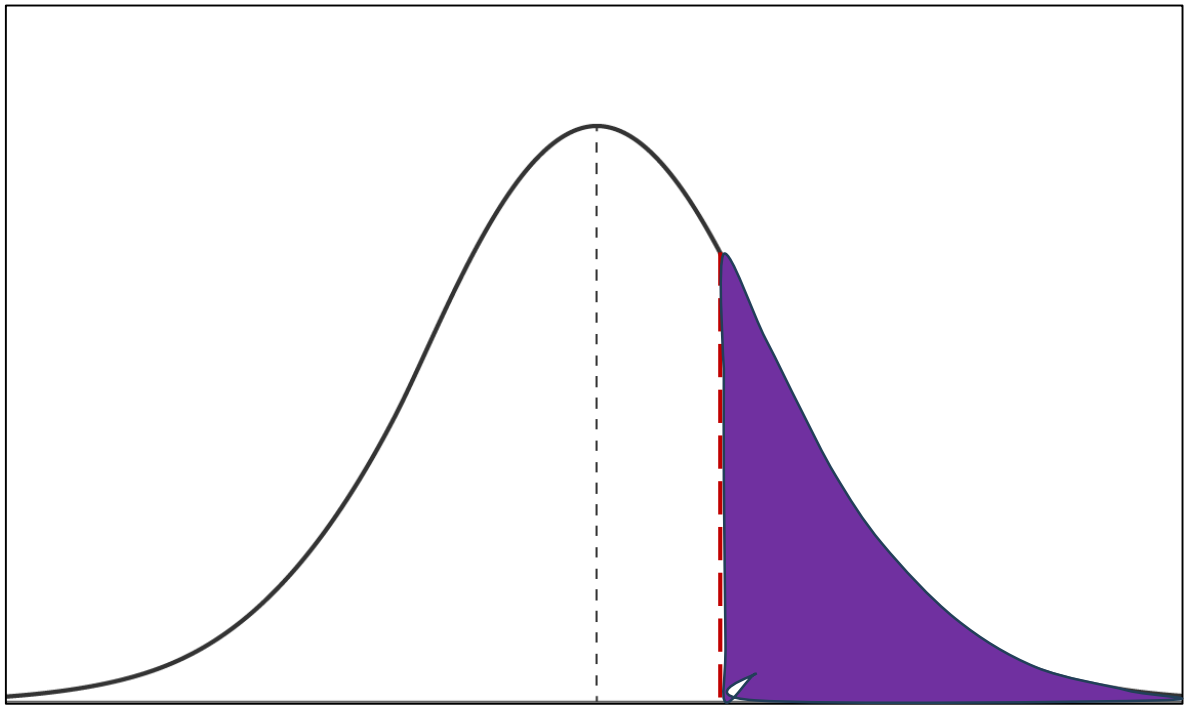
$$Z = \left(\frac{X - \mu}{\sigma}\right)$$

c. Exceed 50 million viewers?

$$P(X > 50)$$

$$1 - P(X \leq 50)$$

$$1 - P\left(\frac{X - \mu}{\sigma} \leq \frac{50 - 60}{7}\right)$$



$$1 - P(Z \leq -1.43)$$

$$1 - 0.0764$$

$$0.9236$$

Example

- The monthly salaries of newly graduated students from BRAC University are normally distributed, with a mean of $250 and a standard deviation of $6. Based on this information find the probability that a randomly selected graduated will get salary:

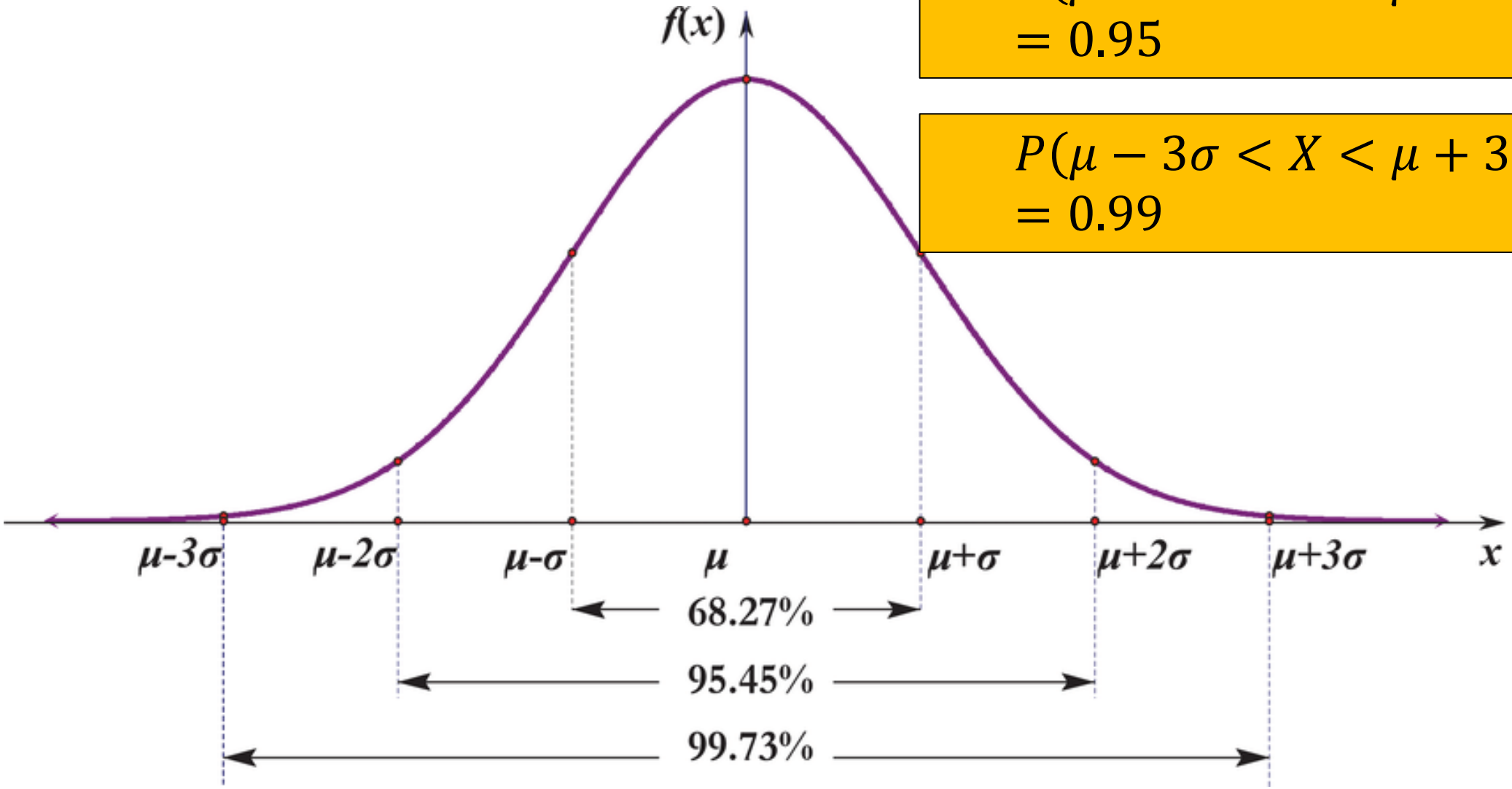
a) At least $235 (Ans: 0.993)
b) Between $255 to $260 (Ans:0.16)
c) At most $300 (Ans:??)

3σ Rules

$$P(\mu - \sigma < X < \mu + \sigma) = 0.68$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.95$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.99$$



Example

- The weight of students follows normal distribution with mean 60 and standard deviation 8 . In what interval would you expect the central 95% of weight to be found?

Here, $\mu = 60, \sigma = 8$

It is expected that 95% of students have an weight between 44 and 76

We know that, $P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.95$

$\mu - 2\sigma = 60 - (2 \times 8) = 44$

$\mu + 2\sigma = 60 + (2 \times 8) = 76$

Example

- For what value of "c", $P(X > c) = 0.01$
- $P\left(\frac{X-\mu}{\sigma} > \frac{c-60}{8}\right) = 0.01$
- $P\left(Z > \frac{c-60}{8}\right) = 0.01$
- $1 - P\left(Z \leq \frac{c-60}{8}\right) = 0.01$
- $P\left(Z \leq \frac{c-60}{8}\right) = 0.99$
- $P(Z \leq 2.33) = 0.99$

$$\frac{c-60}{8} = 2.33$$

$c = ???$

TABLE A

Standard normal probabilities (continued)

z	.00	.01	.02	.03
0.0	.5000	.5040	.5080	.5120
0.1	.5398	.5438	.5478	.5517
0.2	.5793	.5832	.5871	.5910
0.3	.6179	.6217	.6255	.6293
0.4	.6554	.6591	.6628	.6664
0.5	.6915	.6950	.6985	.7019
0.6	.7257	.7291	.7324	.7357
0.7	.7580	.7611	.7642	.7673
0.8	.7881	.7910	.7939	.7967
0.9	.8159	.8186	.8212	.8238
1.0	.8413	.8438	.8461	.8485
1.1	.8643	.8665	.8686	.8708
1.2	.8849	.8869	.8888	.8907
1.3	.9032	.9049	.9066	.9082
1.4	.9192	.9207	.9222	.9236
1.5	.9332	.9345	.9357	.9370
1.6	.9452	.9463	.9474	.9484
1.7	.9554	.9564	.9573	.9582
1.8	.9641	.9649	.9656	.9664
1.9	.9713	.9719	.9726	.9732
2.0	.9772	.9778	.9783	.9788
2.1	.9821	.9826	.9830	.9834
2.2	.9861	.9864	.9868	.9871
2.3	.9893	.9896	.9898	.9901
2.4	.9918	.9920	.9922	.9925

Thank you!